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ORIGINAL ARTICLE

Myrobalan-Mediated Nanocolloids in Controlling Marine Pathogens

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Abstract
Aquaculture production is affected by disease outbreak, which affects the production, profitability, and sustainability of the global aquaculture industry. Antibiotics have been widely used to control various infectious diseases. Indiscriminate usage of antibiotics results in development of antibiotic resistance in pathogens. This current study aims to synthesize myrobalan-mediated green silver nanocolloids (MBNc) by using the extract of three myrobalans and characterized by using various physiochemical techniques. Antibacterial potential of MBNc was screened in vibriosis causing pathogens (*V. harveyi*, *V. alginolyticus*, *V. Parahaemolyticus*), and foodborne pathogen *S. haemolyticus*, isolated from infected fish. Further, the presence of ESBL genes including CTX-M-15 and Amp C was analyzed in control and MBNc-treated strains. From our studies, it was observed that MBNc was very effective in controlling the growth. MBNc confirmed the anti-biofilm property in all tested marine pathogens and effectively abolish the genes encoding CTX-M-15 in tested pathogens. Thus, MBNc can be formulated to control the growth of marine pathogens and it can be used as an alternative to antibiotics to prevent infection in cage culturing and aquafarming.

Keywords Myrobalans · Silver nanocolloids · Foodborne pathogens · *V. harveyi* · *V. alginolyticus* · *V. parahaemolyticus* · *S. haemolyticus*

Introduction

Seafood is a wonderful source of protein, all the essential vitamins, minerals, and most importantly omega 3 fatty acids which are available only through seafood [1]. According to the statistics, 151.2 million tonnes of seafood is consumed per year by humans. There is a huge gap in the consumption of food and production. To meet the ever-growing need of seafood, many aquaculture methods have been initiated to grow certain species in a regulated condition to produce about 110.2 million tonnes per year. Seafood

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contributes to nearly 215 billion dollars, accounting for ~0.3% of the world economy [2]. The major concern for both aquaculture and marine fisheries is the maintenance and prevention of many infections which affect their commercial value which greatly affects the economy. Fish infections affect either directly or by affecting their taste and causing associated illnesses in the host which affects the commercial value. Marine fish infections lead to reduction in potential catch due to the decreased number of live fishes available due to increased mortality slow growth, associated bad taste, lower meat quality, decreased shelf life, and associated human health concerns which make the fish products unlikely to be bought [3].

Vibrio harveyi, *Vibrio alginolyticus*, and *Vibrio parahaemolyticus* cause vibriosis especially in the vertebrate marine organisms. Vibriosis is a serious illness associated with marine fisheries and hence communicable to humans by direct consumption of uncooked fishes and shellfishes or rarely by direct exposure to seawater. Due to its pathogenicity in shellfishes, oysters, and other common fishes, infected organisms are undesirable for purchase causing commercial loss. In human consumption or exposure to *Vibrio* resulted in watery diarrhea, stomach cramps, vomiting, fevers and chills, and other problems associated with the gastrointestinal tract and severe digestive problems associated with water loss [4]. Various exotoxins and virulence factors are associated with *Vibrio* infection in fish as well as in human that include proteases, hemolysins, phospholipases, and cytotoxins [5].

V. harveyi is a free-swimming marine aquatic bacterium which is responsible for major infections in marine vertebrates and invertebrates. *V. harveyi* is associated with wound infection and necrotizing infection in humans [6]. *V. alginolyticus* is a highly dominated species of *Vibrio* community, distributed throughout the marine environment. It is a Gram-negative bacterium which grows as a parasite attached to the marine fishes and human beings. *V. alginolyticus* is reported to infect various fishes [7] and causes severe infection in human beings and animals. Infection includes osteomyelitis, intracranial infection, otitis, ocular infections, and peritonitis. Sea foodborne infections include soft-tissue infections resulting in outbreak through food contamination [8]. *V. parahaemolyticus* is an enteropathogen, and survives both in marine and freshwater sources. *V. parahaemolyticus* causes infections in crabs, shrimp, and tail, and fin rot disease in ornamental fishes. *V. parahaemolyticus* is a major causative agent for an ear infection, wound infection, gastroenteritis, and septicemia. The infection of *V. parahaemolyticus* is caused because of the mishandling of seafood during processing and storage and eating uncooked seafood [9]. *S. haemolyticus* is not normal microflora of marine fishes. *Staphylococcus* are commensals present in the skin of humans and animals. The isolation of *Staphylococcus* sp. indicates the improper packing of fish foods and rarely associated with fish diseases. *Staphylococcus* sp. is responsible for foodborne diseases and food poisoning because of the consumption of raw or uncooked food [10].

All the vibriosis-causing pathogens *V. harveyi*, *V. alginolyticus*, and *V. parahaemolyticus*, and foodborne pathogen *S. haemolyticus* adversely affect the marine vertebrates and invertebrates, which ultimately leads to economic loss to aquaculture industries. Improper processing and storage of seafood and uncooked seafood lead to severe life-threatening diseases to human beings. Besides, these pathogens *Vibrio harveyi*, *V. alginolyticus*, *V. parahaemolyticus*, and *S. haemolyticus* were resistant to antibiotics. During the past few decades, due to excessive use of antibiotics in the field of aquaculture, agriculture, and human medicine, these organisms have developed resistance to many antibiotics. Development of multidrug resistance is one of the major impacts developed recently making the antibiotics ineffective [11]. Thus, this is the need of the hour to find the best alternative to protect the whole society through an environmentally friendly nano solution.

Antimicrobial properties of silver nanoparticles, nanocolloids, and nanosuspensions have already been well established and numerous findings have proven its efficacy in treating many multidrug-resistant pathogens such as *Pseudomonas aeruginosa*, *Streptococcus* sp., and ESBL-producing bacteria. Several mechanisms have been identified on how silver nanoparticles interact and eliminate the bacterial infection by interfering certain biochemical pathways [12].

Hence, this research work combined the beneficial effect of three myrobalans of triphala and silver to formulate myrobalan-mediated nanocolloidal system to control marine pathogens, which are adversely affecting the aquaculture industry and human being through its fish-borne pathogens.

Materials and Methods

Pathogenic strains of *V. harveyi* (MP1), *V. parahaemolyticus* (MP2), *V. alginolyticus* (MP3), and *S. haemolyticus* (MP4 and MP5) were obtained from Central Marine and Fisheries Research Institute, Mandapam [13].

Antibiogram Assay

Antibiotic susceptibility of the pathogenic strains was checked by testing the strains of *V. harveyi* (MP1), *V. parahaemolyticus* (MP2), *V. alginolyticus* (MP3), and *S. haemolyticus* (MP4 and MP5) using antibiotic discs such as penicillin 10 µg, ampicillin 2 µg, enrofloxacin 5 µg, gentamycin 30 µg, ceftriaxone 30 µg, streptomycin 10 µg, and cefoperazone sulbactam 75 µg (HiMedia, India). The cultures were grown as per McFarland standard and inoculated on Muller Hinton agar plates and antibiotic discs were added and incubated at 37 °C for 24 h to screen antibiotic susceptibility [14].

Preparation of Triphala Extract and Qualitative Phytocompound Analysis

The aqueous extract of triphala (mixture of myrobalans) was prepared [15–18] and was subjected to qualitative phytochemical analysis [19] and quantitative carbohydrate analysis by anthrone method [20].

Myrobalan-Mediated Nanocolloid Synthesis and Physicochemical Characterization of MBNC

Myrobalan-mediated synthesis of green nanocolloids (MBNc) was carried out as described previously [15]. The noticeable color change was monitored periodically for the synthesis of MBNc. UV–Vis spectroscopy was used to confirm the MBNc synthesis. The spectral scan was carried out between 300 and 700 nm. The maximum SPR peak confirmed the synthesis of MBNc. The synthesized silver nanocolloid was purified by washing the pellet with autoclaved water, followed by centrifugation. Pellet was then resuspended in 10% DMSO and sonicated for 5 min and utilized for further characterization [21–23].

The size of MBNc was determined using Zeta sizer. The particle size was measured based on the light scattering measurement carried out during the dynamic motion of the nanoparticles present in MBNc for 1 min at 25 °C. The graph displays the hydrodynamic diameter and PDI of the nanocolloidal particles. Zeta potential values are measured using the same instrument integrated with Zeta potential analyzer. The shape and size of MBNc were determined using FESEM. EDAX integrated FESEM depicts the elemental composition of nanocolloidal particles [24–26]. HRTEM measurement for MBNc was carried out by using Jeol/JEM 2100 of 200 kV under LaB6 Electron gun with 0.23 nm resolution and lattice resolution 0.14 nm integrated with SAED [24–26].

Antimicrobial Property of Biologically Synthesized Green Nanocolloids (MBNc) Against Marine Pathogens

The antimicrobial potential of MBNc was validated through four different assays, namely, bacteriostatic assay (minimum inhibitory concentration), time-kill curve validating through different concentrations of MBNc, and bactericidal assay (minimum bactericidal concentration) and antibiofilm assay.

The strains of *V. harveyi*, *V. alginolyticus*, and *V. parahaemolyticus*, and 2 different strains of *S. haemolyticus* were revived in LB broth. All the assays were performed at 0.5 McFarland standard.

The preliminary antimicrobial activity of extract of myrobalans and MBNc was analyzed by the agar well diffusion method. The fresh culture was swabbed evenly on LB agar plates and allowed to dry. Then, the wells were punctured using sterile 6 mm tips. Wells were loaded with ampicillin 25 μg, myrobalan extract 50 μl, and 25 μl and 50 μl of MBNc and incubated at 37 °C for a period of 24 h. Zone of inhibition was recorded [21].

Bacteriostatic assay, time-kill curve, bacteriocidal assay, and biofilm assay for MBNc were carried out by using broth dilution method using microtiter plates [27]. All the assays were performed in triplicate to carry out statistical analysis by using a *t*-test to identify the significant difference among ampicillin- and MBNc-treated strains [13, 28].

Genomic DNA was isolated from control, and MBNc- and ampicillin-treated strains of *Vibrio* sp., and 2 strains of *S. haemolyticus*. The genes encoding CTX-M-15 and AmpC were amplified from control and treated samples [29]. The PCR product was run using agarose gel electrophoresis to validate the presence of genes upon treatment with MBNc and ampicillin and compared with the control [30].

Results and Discussion

Antibiotic Susceptibility of Marine Pathogens

Antibiotic susceptibility of the pathogenic strains *V. harveyi* (MP1), *V. parahaemolyticus* (MP2), *V. alginolyticus* (MP3), and *S. haemolyticus* (MP4 and MP5) was tested using antibiotic discs such as penicillin 10 μg, ampicillin 2 μg, enrofloxacin 5 μg, gentamycin 30 μg, ceftriaxone 30 μg, streptomycin 10 μg, and cefoperazone sulbactam 75 μg. From the result, it was identified that MP3, MP4, and MP5 tested organisms showed resistance to penicillin and ampicillin and were susceptible to all other antibiotics tested (Table 1) whereas *V. harveyi* (MP1) strain was resistant to penicillin 10 μg, ampicillin 2 μg, enrofloxacin 5 μg, and ceftriaxone 30 μg.

Table 1 Antibiotic susceptibility of marine pathogenic bacteria

S. No	Strain	Penicillin 10 µg	Ampicillin 2 µg	Enrofloxacin 5 µg	Gentamycin 30 µg	Ceftriax-one 30 µg	Streptomycin 10 µg	Cefoperazone sulbactum 75 µg
1	*V. harveyi* (MP1)	R	R	R	S	R	S	S
2	*V. alginolyticus* (MP2)	S	S	S	S	S	S	S
3	*V. parahaemolyticus* (MP3)	R	R	S	S	S	S	S
4	*S. haemolyticus* (MP4)	R	R	S	S	S	S	S
5	*S. haemolyticus* (MP5)	R	R	S	S	S	S	S

R resistant, *S* susceptible

Phytochemical Analysis of Biological Extract of Myrobalans

Phytocompound-enriched extracts of triphala myrobalans were prepared and concentrated. Phytochemical analysis of biological aqueous extract of triphala myrobalans revealed the presence of carbohydrates, phenols, saponins, terpenoids, flavonoids, alkaloids, terpenoids, anthocyanin, steroids, and carotenoids (Table 2). Medicinal plants produce phenols, flavonoids, terpenoids, and tannins that contribute to their medicinal property. Flavonoids in combination with phenols, polyphenols, saponins, and other tannins exhibit potential antibacterial, anti-tumor, and antiviral properties [31]. Antioxidant and reducing properties of these phytoconstituents help in the reduction of silver nitrate into nanocolloidal silver. The quantitative estimation of carbohydrates confirmed the presence of total carbohydrate (90 mg/ml) of concentrated triphala extract. This confirmed the presence of carbohydrates which may include the presence of all polysaccharides. The cellulose is one of the carbohydrates which can be extracted easily through water and accounts for major proportion in the triphala extract. Previous studies have reported the presence of cellulose as a major biochemical constituent in triphala extract [32].

Biogenesis of Myrobalan-Mediated Green Nanocolloids (MBNc) and Physicochemical Characterization

Myrobalan-mediated green nanocolloid (MBNc) was prepared and initial color change was observed from pale brown that gradually changed into brownish green to dark brown nanocolloidal particles (Fig. 1a). This color change is the preliminary signature which indicates the synthesis of biomolecule-mediated nanoparticle in green colloidal system. At macroscale, the electrons are loosely packed; at nanoscale the electrons are lightly packed with restricted movement, so the intensity of scattering light varies when compared to macroscale and nanoscale. Hence, the nanoscaled particles look dark brown color which is an indicator for the successful synthesis of nanocolloids [33]. Phytoconstituents such as starch, cellulose, phenols, poly phenols, and flavonoids act as reducing and capping agents and aid in the formation of nanoparticles containing colloidal solution. Further, the synthesis of myrobalan-mediated green nanocolloids (MBNc) was further confirmed by physicochemical characterization.

Table 2 Qualitative phytochemical analysis of aqueous extract of myrobalans

Extract	Carbohydrates	Phenol	Tannins	Flavonoids	Saponins	Anthocyanins	Terpenoids	Alkaloids	Steroids	Carotenoids
Myrobalans extract	+	+	+	+	+	+	+	+	+	+

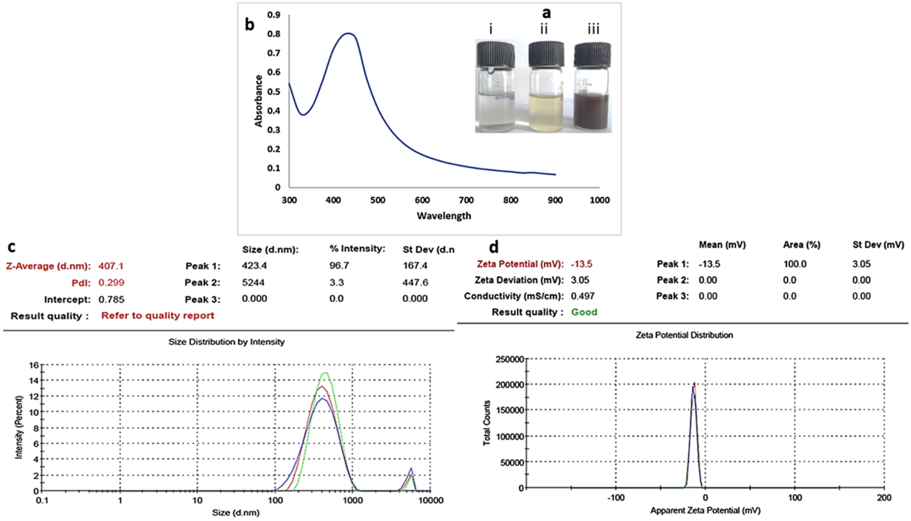

Fig. 1 (Inlet a) (i) 1 mM silver nitrate solution; (ii) extract of myrobalans, when mixed with 1 mM silver nitrate solution; (iii) color change during biological synthesis of green nanocolloid synthesis (MBNc). (b) Uv–Vis absorption spectra of natural polymer–mediated nanocolloid (MBNc). (c) Particle size distribution of MBNc, based on random motion of colloidal particle dispersed in medium. (d) Zeta potential analysis of MBNc

UV–Visible spectrometer scanning was performed between 300 and 800 nm and SPR peak was observed at 430 nm. Silver nanoparticles have strong absorption and scattering of light. When the nanoparticles are exposed to particular wavelength, because of conduction of electrons, they undergo oscillation. The oscillation created by nanoparticle will be much stronger than non-plasmonic particles [33]. The oscillation developed by the nanoparticles will develop a strong peak, which is called as SPR peak. The SPR peak of MBNc was observed around 430 nm which corresponds to nanoparticles which are spherical in shape (Fig. 1b). Myrobalan extract was reported to have phytocompounds, polymers of carbohydrates, polysaccharides, polyphenols, enzymes, coenzymes, and other proteins which act as a bioreductant in reducing the silver ions [24–27].

The size of MBNc was found as 407.1 nm in range, based on Brownian movement of the nanoparticles dispersed in the colloidal solution with the poly dispersive index of 0.299 with intercept of 0.785 (Fig. 1c). The size of MBNc was measured using DLS which was based on random dynamic motion of the particles. PDI value of 0.299 is a highly acceptable value, described as homogenous. The Zeta potential value of MBNc was found as − 13.5 mV (Fig. 1d). The negative value implies that the negative surface charge of − 13.9 will create a repulsive force between the particles and prevent agglomeration in the dispersed medium. The size, charge, and aggregation nature of the MBNc mainly depends on natural polymers and phytocompounds present in myrobalan extract. These phytocompounds and biological macromolecules uniformly laid over the surface of nanoparticles protect as a cap, to prevent the agglomeration. Thus, these biomolecules work synergistically along with silver in imparting effective antibacterial activity [21–23].

FESEM images revealed that the nanocolloids were embedded with spherical-shaped nanoparticles at 1 μm scale (Fig. 2a). The mixture of three myrobalans may produce these spherical-shaped nanoparticles, which may have its unique role in targeting the bacterial cells through its shapes that helps in penetrating through the bacterial cell wall. EDAX analysis revealed the presence of silver at 3 keV indicates the presence of silver (Fig. 2b). The histogram of EDAX analysis depicts the elements present in MBNc (Fig. 2c). The presence of cl works synergistically with silver in enhancing its antimicrobial activity [34, 35].

HRTEM analysis showed the distribution, average size, and shape of the MBNc. HRTEM images of MBNc revealed its polydispersed and spherical (Fig. 3a) nature. The average size of MBNc was found as 23.63 nm. SAED patterns are signature for the crystals. HRTEM analysis showed the distribution, average size and shape of the MBNc. HRTEM images of MBNc revealed its spherical nature and polydispersity (Fig. 3a). The average size of MBNc was found as 23.63 nm. SAED patterns are the signature for the MBNc, which showed the spots which indicated the polycrystalline nature of MBNc (Fig. 3b) [29].

Antibacterial Activity of MBNc

Detection of Antibacterial Activity of MBNc Through Agar Well Diffusion Assay

The preliminary antibacterial effect of MBNc was evaluated against *V. harveyi* (MP1), *V. parahaemolyticus* (MP2), *V. alginolyticus* (MP3), and *S. haemolyticus* (MP4 and MP5). At 50 μg concentration of MBNc, maximum zone of inhibition was observed as 26 mm, 25 mm, 26 mm, and 20 mm for *V. harveyi* (MP1), *V. parahaemolyticus* (MP2), *V. alginolyticus* (MP3), and *S. haemolyticus* (MP4 and MP5) respectively (Table 3). MBNc when loaded into the wells, diffuses into the maximum area. One of the properties of nanoparticles is large surface area where nanoparticles encounter microorganism in maximum area, which inhibits and kills the organism through several mechanisms

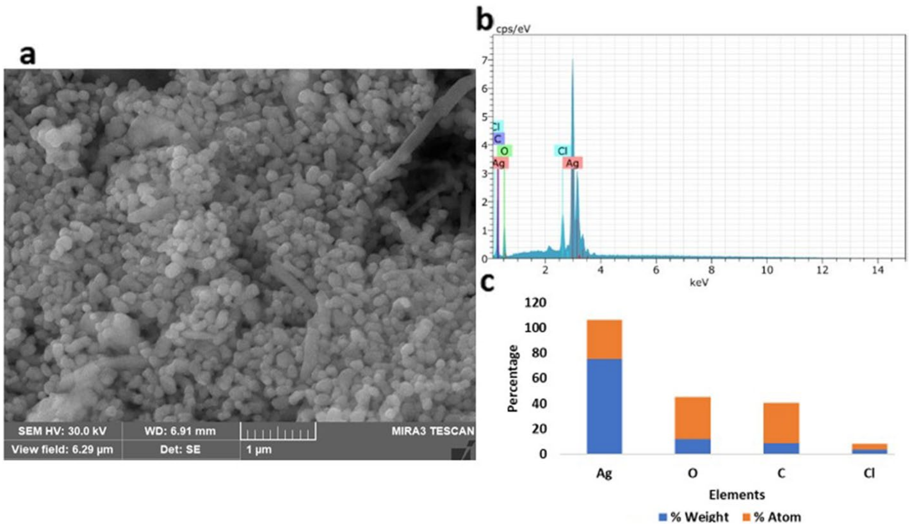

Fig. 2 (a) Field emission scanning microscopic image of MBNc at 1 μm scale. (b) Qualitative and quantitative analysis of elements present in MBNc obtained through EDAX spectrum. (c) Histogram showing the percentage atom and percentage weight of elements present in MBNc

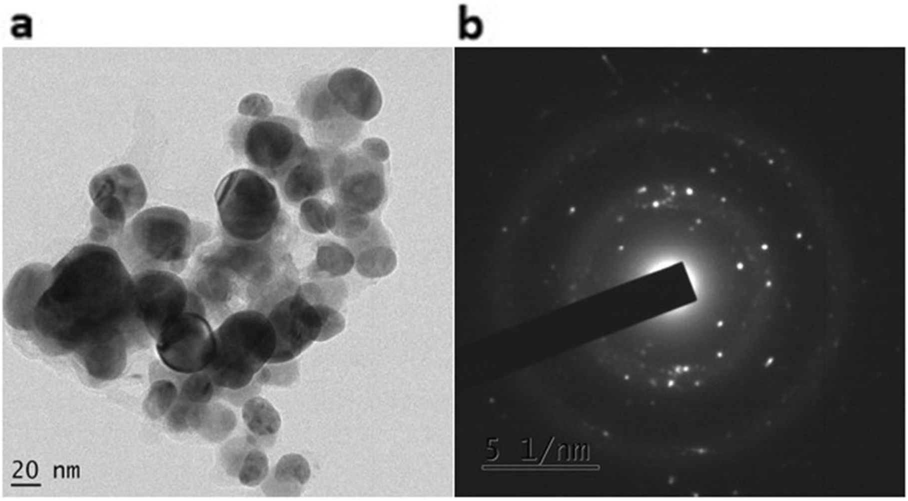

Fig. 3 (a) HRTEM analysis of MBNc at 20 nm scale. (b) SAED pattern of MBNc shows polycrystalline nature of MBNc

Table 3 Measurement of zone of inhibition upon treatment with MBNc against antibiotic resistant marine pathogens

S. No	Strains	Zone of inhibition in mm		Ampicillin 25 µg
		MBNc (50 µg)	MBNc (25 µg)	
1	*V. harveyi* (MP1)	26	20	No zone
2	*V. alginolyticus* (MP2)	25	22	No zone
3	*V. parahaemolyticus* (MP3)	26	20	28
4	*S. haemolyticus* (MP4)	20	18	No zone
5	*S. haemolyticus* (MP5)	20	18	No zone

[36]. The synergistic effect of phytocompounds and silver enhances the antibacterial potential of MBNc.

Detecting the Bacteriostatic and Bactericidal Concentration of MBNc

MBNc was validated to detect the concentration at which it completely inhibits the visible growth of organism. The minimum inhibitory concentration for *V. harveyi* (MP1), *V. parahaemolyticus* (MP2), and *V. alginolyticus* (MP3) was found as 6.25 µg, 1.56 µg, and 3.125 µg respectively, and for *S. haemolyticus* (MP4 and MP5) bacteriostatic concentration was found as 12.5 µg. From this result it was observed that each strain responds differently to MBNc. The cell wall of Gram-negative *Vibrio* sp. requires maximum 6.125 µg of MBNc when compared to Gram-positive *Staphylococcus haemolyticus*, which requires 12.5 µg of MBNc (Table 4). Because the cell wall of Gram-positive strains will be thicker than Gram-negative strains, the mode of action of MBNc will require higher concentration to penetrate the cell wall. The bactericidal concentration

Table 4 Determination of bacteriostatic concentration and bactericidal concentration of MBNc against marine pathogens

S. No	Strains	(MBNc) (μg/ml)	
		Bacteriostatic concentration	Bactericidal concentration
1	*V. harveyi* (MP1)	6.25	12.5
2	*V. alginolyticus* (MP2)	3.125	6.25
3	*V. parahaemolyticus* (MP3)	1.56	3.125
4	*S. haemolyticus* (MP4)	12.5	12.5
5	*S. haemolyticus* (MP5)	12.5	12.5

of MBNc was found as 12.5 μg, 3.125 μg, and 6.25 μg for MP1, MP2, and MP3 respectively and 12.5 μg for other 2 strains of *Staphylococcus haemolyticus*. From the result it was found that MBNc has potent bacteriostatic and bactericidal property against antibiotic-resistant organisms, even at the lower concentration (12.5 μg/ml) (Table 4). Because nanocolloids target the bacterial cell through several mechanisms, the first target is the cell membrane. When the nanoparticles interact with the bacterial cell, it creates pits on the surface of cell membrane and disrupts membrane permeability. Then, the silver ions capped with biomolecules enter the cell through porins, which interfere with the normal functions of the plasma membrane [37]. Previous reports are supporting the functioning of nanoparticles in such a way that imbalance of pH gradient will be created which induces the leakage of H^+ ions in *Vibrio* sp. [38]. MBNc may interact with thiol group of many functional proteins and enzymes, which may ultimately inhibit the respiratory chain and block the synthesis of ATP. There is an evidence that silver nanoparticles intercalate with phosphorus and sulfur group of DNA and inhibit the replication of the cell. Further the interaction of Ag ions with signalling pathway may alter the tyrosine phosphorylation, which leads to the adverse effect on mitosis and meiosis and finally leads to cell death. Silver ions also generate more and more ROS inside the cell, which is very toxic and leads to cell death [39]. Our results suggested that the physical property such as size, shape, and charge and chemical properties such as interaction within the bacterial cell of MBNc played a crucial role to exert its action as a potent bacteriostatic and bactericidal agent.

MBNc Modulates Growth Kinetics in Marine Pathogens

The dose-dependent activity of MBNc was observed in all the test strains *V. harveyi* (MP1), *V. parahaemolyticus* (MP2), *V. alginolyticus* (MP3), and *S. haemolyticus* (MP4 and MP5) after 12 h of treatment (Fig. 4a). To validate the effect of MBNc on growth, different concentrations are used and growth rate was measured at different time intervals and with different concentrations of MBNc (1.5625 μg, 3.125 μg, 6.25 μg, and 12.5 μg). At 1.5625 μg the growth rate was decreased by 72%, 81%, 67.5%, 76%, and 61.5% in MP1 to MP5 strains, respectively. At 3.125 μg the growth rate was decreased by 73.8%, 89.8%, 88.1%, 78.9%, and 63.3% in MP1 to MP5 strains, respectively. At 6.25 μg the growth rate was decreased by 90.4%, 93.9%, 93.9%, 86.3%, and 80.8% in MP1 to MP5 strains, respectively. At the concentration of 12.5 μg the growth rate was decreased by 94.88%, 94.69%, 94.3%, 99.0%, and 93.7% in MP1 to MP5 strains, respectively.

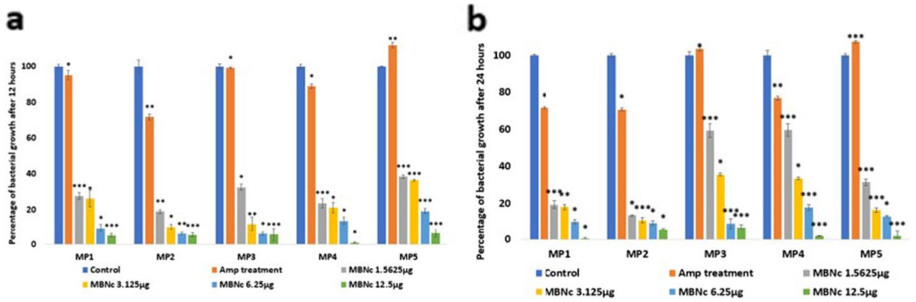

Fig. 4 (a) Percentage of bacterial growth rate after 12 h upon MBNc treatment. (b) Percentage of bacterial growth rate after 24 h upon MBNc treatment

At 24 h the growth rate was further decreased. At the concentration of 1.5625 μg, the growth rate was decreased by 80.6%, 86.6%, 40.9%, 40.3%, and 68.5% in MP1 to MP5 strains, respectively. At the concentration of 3.125 μg, the growth rate was decreased by 82%, 89%, 64.4%, 66.5%, and 83.7% in MP1 to MP5 strains, respectively. At the concentration of 6.25 μg, the growth rate was decreased by 90%, 90.8%, 91.0%, 82.2%, and 87% in MP1 to MP5 strains, respectively. At the concentration of 12.5 μg, the growth rate was decreased by 99%, 94%, 93%, 98%, and 98% in MP1 to MP5 strains, respectively. At the concentration of 12.5 μg, the growth rates of all the tested strains of *Vibrio* sp. (MP1, MP2, MP3) and *S. haemolyticus* (MP4 and MP5) were effectively inhibited even after 24 h of incubation without any further treatment (Fig. 4b). These results suggested that MBNc could be used as potent alternate to control the growth of marine pathogens in marine fishes growing in caged farming.

Effect of MBNc on Biofilm Formation

Biofilm produced by the bacteria is a virulence mechanism which in turn become more antibiotic resistant strain. The hydrophobic nature of bacterial cell communicates and interact by means of signalling molecules to develop biofilms on the surface. In general, antibiofilm agents will disrupt the communication through cell membrane or will quench the signals from bacterial cells that can inhibit the biofilm formation. To check this hypothesis, MBNc was validated to check the anti-biofilm action on the following marine pathogens *V. harveyi* (MP1), V. *parahaemolyticus* (MP2), *V. alginolyticus* (MP3) and *S. haemolyticus* (MP4 and MP5). From the antibiofilm assay it was found that MBNc inhibited the biofilm formation in a dose dependent manner of four different concentrations 1.5625 μg, 3.125 μg, 6.25 μg and 12.5 μg (Fig. 5a). At the concentration of 1.5625μg the biofilm formation was inhibited up to 82%, 82%, 54%, 77%, and 78.9% for MP1 to MP5, respectively. MBNc at 3.125 μg of concentration, effectively inhibited the biofilm formation up to 90%, 85%, 86.5%, 90.9%, and 83.6% for MP1 to MP5, respectively. At the concentration of 6.25 μg the biofilm formation was inhibited up to 92%, 96.8%, 95.6%, 92%, and 88.9% for MP1 to MP5, respectively. At the concentration of 12.5 μg the biofilm formation was inhibited maximum up to 97.9%, 99.7%, 97.8%, 94.9%, and 90.8% for MP1 to MP5, respectively.

As per the bacteriostatic and bactericidal assay, the MBNc was effective at the concentration of 12.5 μg, supporting to these findings the MBNc also showed the maximum biofilm inhibition potential of 99%. It was reported that smaller the size of nanoparticle

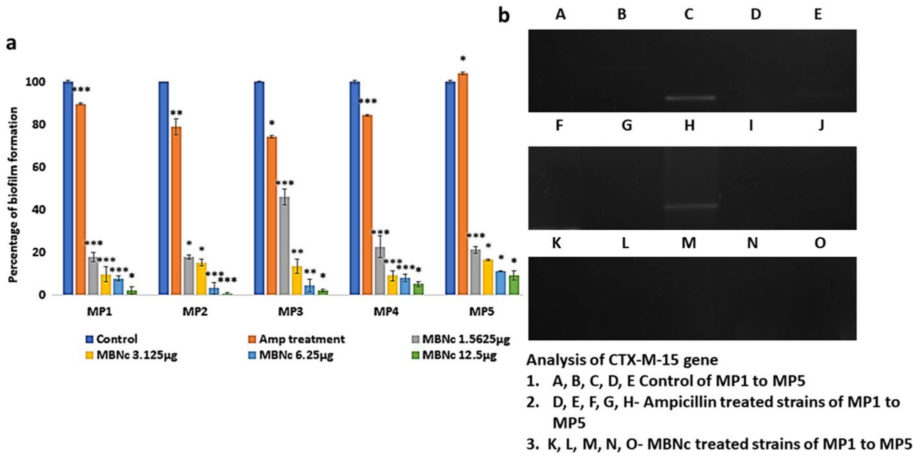

Fig. 5 (a) Percentage of biofilm formation in marine pathogenic strains upon treatment with different concentrations of MBNc. (b) Analysis of CTX-M-15 gene through PCR amplification using gene specific primers: wells A, B, C, D, E—control (without MBNc treatment) of *V. harveyi* (MP1), *V. parahaemolyticus* (MP2), *V. alginolyticus* (MP3), *S. haemolyticus* (MP4), *S. haemolyticus* (MP5), whereas only MP3 strain showed the presence CTX-M-15 gene. Wells F, G, H, I, J are ampicillin-treated strains, where there is downregulation of CTX-M-15 gene in MP3 strain alone. Wells K, L, M, N, O—treated with MBNc, where in well M, MP3 showed complete abolishment of CTX-M-15 gene in *V. alginolyticus* (MP3) strain

greater the antimicrobial effect, according to HRTEM the average size of particles was found around 23.63nm. Hence, MBNc easily penetrate the cell membrane and disrupt the surface of bacterial cell. The quorum sensors are the molecules secreted out for cell to cell communication. The secretion of these signalling compounds could have been inhibited by the nanocolloids, through inhibiting the synthesis of signalling molecules, degrading, or inactivating the signalling enzymes, analogs, disrupting the signalling cascade. MBNc can effectively disrupt the cell membrane, targets the central dogma of cell by disturbing whole mechanism of replication, transcription, and translation, sequentially all the signalling cascade will be disturbed [42]. There were previous reports supporting that nanoparticles controlling the biofilm formation in organism such as *S. dysentriae*, *V. parahaemolyticus* and *S. infestis* through interfering with cell to cell adhesion [43]. Our results suggest that MBNc can be effective anti-biofilm agent, even at lower concentration, which could be used as an effective alternative to antibiotics to control the infection caused by marine pathogen and marine associated food borne pathogens. Thus, this can be effectively used to control the usage of antibiotics in marine cage culturing and in aqua culturing industries.

Effect of MBNc on Antibiotic Resistant Gene

Through gene amplification studies it was observed that AmpC gene was not present in all the tested marine pathogens. However, CTX-M-15 expression was found only in *V. alginolyticus* (MP3). Absence of CTX-M-15 gene in *Vibrio* sp (MP1, MP2, MP3) and *Staphylococcus haemolyticus* (MP4 and MP5) suggested that virulence and antibiotic resistance were governed by some other ESBL genes present in these strains [44]. From the gel documentation, it was found that in control (without treatment), CTX-M-15 gene was observed in MP3 (*V. alginolyticus*). Upon treatment with ampicillin, CTX-M-15 gene was present, however MBNc treatment (5µg and 10µg), completely abolished the CTX-M-15 gene. This demonstrates that MBNc may efficiently target the transcription and translational level (Fig. 5b). These results suggest that MBNc can effectively target on all strains with and without the expression of ESBL gene.

Conclusion

The current research is focused on myrobalan-mediated synthesis of green nanocolloids by using three myrobalans *Emblica officinalis* (Indian gooseberry), *Terminalia belerica* (Belliric Myrobalan), and *Terminalia chebula* (Chebulic Myrobalan), commonly called as triphala, which is widely used in ayurvedic medicine which is of great medicinal value. MBNc showed effective antibacterial and antibiofilm activity even at low concentrations. Further, MBNc abolished CTX-M-15 gene in MP3 (*V. alginolyticus*). This phytocompound-enriched biological extract of three myrobalans, mediated nanocolloids could be the potent alternative to prevent the indiscriminate usage of antibiotics in aquaculture and fish farming. The usage of MBNc also prevents the spread of the antibiotic-resistant organisms to humans who are consuming the marine foods. MBNc was formulated using traditional ayurvedic medicine, which does not cause any adverse effects when it is being used at moderate level. Further development and validation of MBNc through various formulations as nanospray, nanovaccine, and nano feed additive are needed to prevent the pathogenic infection in aqua culturing of many economically and commercially important species such as penaeid shrimp, gilthead sea bream, European seabass, tiger puffer, groupers, Asian seabass, and Cobia.

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Author Contribution SH conceived and designed research. SR and PP conducted experiments. PR contributed pathogenic strains. SH and PR analyzed data. All authors wrote the manuscript. All authors read and approved the manuscript.

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Code Availability Not applicable.

Declarations

Conflict of Interest The authors declare no competing interests.

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